

Analytical Optimisation of Backscatter Reliability Using Packet-Level Error-Control Coding and Interleaving

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1. Introduction

Our group project investigated reliability in a backscatter communication system consisting of a carrier generator, a backscatter tag, and a receiver. The tag does not transmit actively; it reflects and modulates the carrier, so the signal arriving at the receiver is shaped by two cascaded links - the carrier-to-tag path and the tag-to-receiver path. This dual-link dependence makes the received signal weak and strongly dependent on tag position, which framed the whole project.

We carried out two experimental optimisations, both at the physical layer. The first varied the baud rate (100k down to 50k); 80k gave the best bit error rate (BER) at the reachable edge positions, while 70k was the only configuration that received any packets at the midpoint P3, although that link was marginal (BER $\approx 30\%$ and a long experiment duration). The second pushed the baud rate up to 250k with longer packets, attempting to exploit brief windows of low interference; this degraded reliability everywhere except close to the carrier (P1), where the packet reception rate (PRR) collapsed to single-digit percentages at the other positions.

Both optimisations were bounded by the same constraint: they only adjusted timing parameters. Neither touched how the payload itself was protected against errors. This analytical optimisation steps beyond that hardware/timing boundary and asks a coding-layer question instead - what would be expected to happen if the payload carried its own error-correction capability. No additional experiments or simulations were performed; the reasoning below builds on our measured results and on communication theory.

2. Identified Limitation

In our experimental work the only mechanism gating a packet was the sync word, which determines whether a packet is *accepted* by the receiver. The sync word does not protect the payload. Once a packet was accepted, any bit errors inside it remained - there was no redundancy, no parity, and no way to recover a corrupted bit without a full retransmission, which the system did not perform.

This is a meaningful limitation precisely because of what our data showed. At the edge positions, packets were received but still carried non-trivial BER (for example, around 10–18% at baseline, and still several percent even at the better 80k configuration). These are exactly the conditions where a payload is delivered but partially wrong. Separately, the Optimisation 2 results exposed bursty, clustered packet behaviour: reception came in short bursts during low-interference windows rather than as a steady stream. So the system suffered from *two distinct failure modes* - bit errors inside received packets, and clustered loss/corruption tied to interference - and neither was addressed by tuning baud rate or packet length. The coding layer was

left entirely unoptimised.

3. Hypothesis

If lightweight forward error correction (FEC) and bit interleaving were added on top of the best baud-rate configurations, then the delivered payload reliability would be expected to improve at tag positions where packets are received but contain bit errors. Interleaving would be expected to disperse burst errors across multiple codewords, so that the per-codeword error count stays within what the FEC can correct, rather than overwhelming a single codeword. The improvement would be expected to be strongest at the moderate-BER edge positions, while the midpoint P3 would be expected to remain difficult, because coding cannot recover a packet that is never received or detected in the first place. This gain would come at the cost of reduced useful throughput, increased latency, and additional processing and memory on the constrained tag.

4. Theoretical Motivation

4.1 The backscatter dual-link constraint

Because the backscatter tag does not generate its own RF signal, the usable signal power depends on the product of the two link budgets rather than a single transmitter-to-receiver path. Prior backscatter work has emphasised that keeping the carrier close to the tag and using narrow-band backscatter transmissions improves effective receiver sensitivity and communication range [1]. This is consistent with our own observation that positions nearer the carrier (P1) consistently performed best and that the midpoint, where both links are simultaneously weak, was the hardest. The dual-link fragility is the root reason BER is high enough to matter at the packet level.

4.2 Symbol duration and the receiver decision window

The baud rate f_{baud} sets the symbol duration $T_s = 1/f_{\text{baud}}$. A lower baud rate lengthens T_s , giving the receiver a longer observation window per symbol and more energy to integrate before making a decision, which lowers the probability of a symbol (and hence bit) error under weak or noisy conditions [2]. This is the mechanism behind our Optimisation 1 result, and it also explains why the 250k configuration was fragile: a very short T_s leaves almost no margin for detection once the signal is weak. Coding does not change T_s ; it operates on top of whatever raw bit-error probability this window produces.

4.3 From bit errors to packet errors

For an uncoded packet of N bits with independent bit-error probability p , the probability that the packet contains at least one error is

$$P_{\text{packet error}} = 1 - (1 - p)^N. \quad (1)$$

Equation (1) shows why even a modest p produces poor payload reliability, and why it gets worse as N grows - which is directly relevant to Optimisation 2, where the packet length was increased. With a 12-byte payload ($N = 96$ bits) and, say, $p = 0.05$, the packet-error probability is already very high. This is the quantitative argument for adding redundancy rather than simply sending more or longer raw packets.

4.4 Forward error correction

FEC adds structured redundancy so the receiver can correct a bounded number of bit errors per codeword without retransmission [2]. Effectively it lowers the *delivered* payload error rate at the

expense of raw throughput, because some of the transmitted bits now carry parity rather than data. For a constrained backscatter tag, the relevant codes are lightweight ones (for example simple block codes) rather than computationally heavy schemes, since encoding must run within the processing and memory constraints of the embedded backscatter platform.

4.5 Interleaving and burst errors

Equation (1) and standard FEC analysis both assume errors are *independent*. Real wireless channels, including our own Optimisation 2 data, show that errors are often *bursty*: they arrive in clusters tied to fading or interference. A burst can exceed the correcting capacity of a single codeword even when the average error rate is low. Interleaving reorders the coded bits before transmission and restores their order at the receiver, so that a burst on the channel is spread across many codewords, each of which then sees only a few errors [3]. In other words, interleaving makes a bursty channel look closer to the independent-error channel that FEC is designed for. This is why the two techniques are proposed together rather than separately: in our specific interference-heavy environment, FEC alone would be undermined by the burst behaviour we actually measured.

4.6 What coding cannot do

Coding and interleaving act only on packets that are received and accepted. They cannot recover a packet that is never detected - for instance at P3, where the dominant problem was reception failure, not payload corruption. This boundary matters for honesty: the proposed optimisation targets the “received-but-corrupted” regime, not the “not-received-at-all” regime.

5. Expected Behaviour Based on Our Results

Mapping the theory onto our measured configurations:

80k baud (edge-reliability candidate). This configuration already minimised BER at the reachable positions, so it represents the best raw substrate for coding. Adding FEC with interleaving here would be expected to push the *delivered* payload error rate lower still, since the residual bit errors are lower than in the baseline and high-rate cases, making this configuration a more plausible candidate for lightweight coding. However, the code strength would still need to be chosen carefully, because a very light code may not be sufficient at the higher-BER edge positions. This is where the largest practical benefit would be expected.

70k baud (coverage candidate). This was the only configuration to reach P3, but marginally. Coding might modestly improve the *quality* of the few packets that do arrive at P3, but it cannot increase how many arrive, so the midpoint would be expected to remain weak. At 70k’s edge positions, the same edge-position reasoning as 80k applies.

250k with longer packets (negative example). Our data showed that this was not a useful direction under our experimental conditions: short symbols plus long packets maximise exposure to interference, and by Equation (1) a larger N inflates packet-error probability. Coding would not be expected to rescue this configuration by itself, because the underlying raw error rate and burstiness are too severe; the redundancy overhead would be spent fighting a channel that is too unreliable at that operating point. This reinforces that coding complements a sensible baud-rate choice rather than substituting for one.

Received vs. lost packets. Across all positions, the consistent theme is that coding helps the corrupted-but-received packets and does nothing for the lost ones. The expected reliability gain is therefore position-dependent and tracks where packets are actually arriving.

6. Tradeoff Analysis

The proposed optimisation has clear costs. **Useful throughput** drops because parity bits displace data bits; for a tag already operating at tens of bytes per second, this is a real reduction. **Latency** increases, both from the interleaver (which must buffer a block of bits before it can be transmitted or deinterleaved) and from the longer effective transmission needed to carry the redundancy. **Complexity** rises on the tag side - encoding and interleaving require additional logic and memory, which is a meaningful constraint for a constrained embedded backscatter tag/platform. There is also an interaction with packet length: stronger codes and deeper interleavers generally want larger blocks, yet Equation (1) and our Optimisation 2 result both warn against long raw packets in this channel. The design would therefore have to balance interleaver depth against exposure, which is itself a non-trivial tradeoff.

The net judgement is a classic reliability-versus-efficiency exchange: better delivered reliability at moderate-BER positions, paid for in throughput, latency, and tag complexity.

7. Conclusion

Packet-level FEC with interleaving is a meaningful analytical optimisation for this project because it extends the work from physical-layer parameter tuning into communication-system design. Our experimental optimisations established *where* the system can deliver packets and at what raw BER; the coding layer addresses *what happens to those packets once delivered*, which the baud-rate experiments left untouched. Crucially, it complements rather than replaces baud-rate optimisation: a sensible baud rate, such as 80k, provides a lower raw error rate where lightweight coding is more likely to be useful, while interleaving handles the burstiness our Optimisation 2 data revealed. It is not a cure for the midpoint, where reception itself fails, and that limitation is part of why the proposal is framed cautiously. Within its proper scope - received-but-corrupted packets at reachable positions - it would be expected to improve delivered reliability, at a well-defined cost in throughput, latency, and complexity.

AI use statement

AI was used only for language improvement and structuring support. The technical reasoning, project data, and conclusions are based on the group's own experimental reports and communication theory.

References

- [1] A. Varshney, O. Harms, C. Pérez-Penichet, C. Rohner, F. Hermans, and T. Voigt, "LoRea: A Backscatter Architecture that Achieves a Long Communication Range," in *Proc. 15th ACM Conf. on Embedded Network Sensor Systems (SenSys '17)*, Delft, Netherlands, 2017, pp. 18:1–18:14. DOI: 10.1145/3131672.3131691.
- [2] B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2001.
- [3] Y. Q. Shi, X. M. Zhang, Z.-C. Ni, and N. Ansari, "Interleaving for combating bursts of errors," *IEEE Circuits and Systems Magazine*, vol. 4, no. 1, pp. 29–42, 2004. DOI: 10.1109/MCAS.2004.1286985.